

Rewarding the Graph Behind the Chain: Topology-Aware Process Supervision for RL and Reasoning Distillation

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Abstract

Long chain-of-thought reasoning in LLMs is neither strictly correct-or-wrong nor purely linear. Outcome-only rewards overlook unjustified intermediate steps, while existing process reward models depend on step-level annotations and fail to capture the implicit conditional dependencies within a reasoning chain. We propose TopoPRM, a topology-aware implicit process reward model that extracts a dependency DAG from each reasoning trace and integrates conditional dependency and sequential continuity as complementary components of a multi-source reward signal. Building on TopoPRM, we design TGSD, a post-training framework that combines supervised warm-start, GRPO optimization under a correctness-first hierarchical reward, and topology-guided on-policy distillation. Across multiple public benchmarks, our method improves average accuracy over standard GRPO at both 7B and 9B scales while generating up to 24% fewer tokens, with consistent gains on elementary and general-domain reasoning.

1 Introduction

Reinforcement learning with verifiable rewards (RLVR) has substantially advanced long chain-of-thought (CoT) reasoning in large language models and now serves as a core mechanism for training mathematical and symbolic reasoning ability (Wei et al., 2022; Kojima et al., 2022; DeepSeek-AI, 2025; Yeo et al., 2025; Shao et al., 2024; Ke et al., 2026). From sparse final-answer rewards (Cobbe et al., 2021; DeepSeek-AI, 2025) to process reward models (PRMs) that provide intermediate step-level supervision for improved reliability (Lightman et al., 2023; Wang et al., 2024; Uesato et al., 2022), current approaches uniformly treat a reasoning trace as a *linear sequence of steps*. In practice, long reasoning is rarely linear: in mathematical derivations, a later conclusion often depends on several non-adjacent premises, an intermediate

equation is reused by multiple downstream derivations, and subgoals branch before merging into the final answer (Besta et al., 2024). Reconciling this structural mismatch has emerged as a core challenge for advancing reasoning supervision beyond linear credit assignment.

Recent reasoning models have made the macroscopic structure of inference clearer and more reliable, yet they remain confined to the CoT-as-result view. DeepSeek-R1 (DeepSeek-AI, 2025), DAPO (Yu et al., 2025), and Tulu 3 (Lambert et al., 2024) demonstrated that outcome-only rewards scale, but they could not distinguish a well-supported derivation from a trace that reached the same answer through unsupported intermediate claims, and they remained vulnerable to reward hacking (Gao et al., 2023; Skalse et al., 2022). On the PRM side, denser step-level signals have been obtained through Monte Carlo rollouts (Wang et al., 2024; Luo et al., 2024), trained verifiers (Zhang et al., 2025; Khalifa et al., 2025; Zhao et al., 2026), and implicit outcome-derived scoring (Yuan et al., 2025; She et al., 2025), yet all treat the reasoning trace as a **linear sequence of steps** and assign step-local credit, without modeling the **non-sequential dependency structure** in which a single intermediate result may support, or be supported by, several non-adjacent steps (Wang et al., 2025). Token-level distillation (Agarwal et al., 2024) inherited the same linear bias by matching teacher distributions step by step.

We propose TOPOPRM, which supervises reasoning through its *implicit topological dependency structure* rather than its surface order (Figure 1). The core idea is to extract, from any free-form reasoning trace, a DAG-shaped topology whose nodes are reasoning steps and whose edges encode surface-evidenced support relations such as expression reuse, variable propagation, claim reference, and explicit step citation. Two technical components turn this idea into a usable signal.

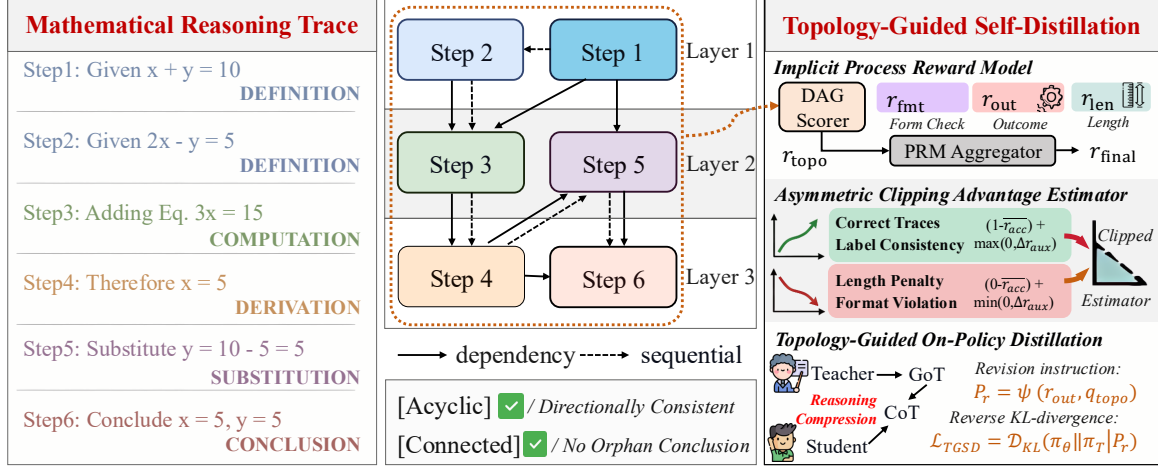


Figure 1: The motivation of TopoPRM. A free-form reasoning trace is first turned into an *Implicit Dependency DAG* with typed nodes and edges, which is then scored to drive training stages with topology-aware diagnostics.

First, a hybrid DAG scorer (rule-based patterns augmented with LLM-based inference) produces dual diagnostics that respectively measure *global topology quality*, including orphan support, direction consistency, and acyclicity, and *local continuity quality*, exposing the unsupported conclusions and redundant branches that final-answer rewards cannot observe. Second, we introduce TGSD, a post-training framework that shares the same topology diagnostics and hierarchically aggregates multi-source reward signals; at inference, it requires no human step labels, no separate verifier, and no graph-format generation. Through three stages of post-training, namely SFT warm-start, GRPO driven by a correctness-first hierarchical reward, and topology-guided on-policy self-distillation, TGSD yields a stronger and more token-efficient reasoning model. Our contributions are threefold:

- Reasoning topology as process supervision.** We formulate long CoT traces as implicit dependency DAGs and define topology-aware process signals that jointly score whether intermediate conclusions are *supported*, *directionally consistent*, and *locally traceable*, capturing failure modes that outcome-only and step-isolated rewards both miss.
- A unified reward and post-training framework.** We instantiate the formulation as TOPOPRM, an implicit PRM driven by topology DAG diagnostics, and embed it into TGSD, which couples SFT warm-start, hierarchical-reward GRPO, and topology-conditioned on-policy distillation through a single DAG Scorer.

- Empirical gains in accuracy and efficiency.** On multiple public math benchmarks at the 7B and 9B scales, TGSD consistently surpasses standard GRPO while compressing max response length by up to 24%, and ablations confirm that topology-aware rewards offer supervision complementary to final-answer signals.

2 Related Work

Reinforcement Learning with Verifiable Rewards. RLVR replaces preference models with programmatic correctness checks on tasks whose answers are objectively verifiable (Shao et al., 2024; DeepSeek-AI, 2025; Lambert et al., 2024; Zeng et al., 2023). Recent work refines GRPO-style optimization through dynamic sampling and clipping (Yu et al., 2025), extended training schedules (Liu et al., 2025), finer credit assignment (Kazemnejad et al., 2025), asymmetric handling of negative samples (Zhu et al., 2025), entropy regularization (Cui et al., 2025), and simplified baselines (Ahmadian et al., 2024; Hu, 2025). Outcome-only signals nevertheless remain structurally blind: traces with very different dependency support can be reward-equivalent (Chen et al., 2026), and reward hacking persists (Gao et al., 2023; Skalse et al., 2022). *TopoPRM injects deterministic topology and continuity diagnostics from latent reasoning DAGs, targeting structural credit assignment directly rather than through learned reward surrogates.*

Process Reward Models for Reasoning. PRMs score intermediate steps to address the sparsity of outcome rewards, originally relying on dense

human annotation (Uesato et al., 2022; Lightman et al., 2023). Subsequent work reduces labeling cost via Monte Carlo or tree-search rollouts (Wang et al., 2024; Luo et al., 2024; Zhang et al., 2024; Lu et al., 2024; Zhang et al., 2025; Guan et al., 2025), generative PRMs make verifier reasoning explicit (Khalifa et al., 2025; Zhao et al., 2026; She et al., 2025), and process advantage verifiers tie step values to rollout-induced returns (Setlur et al., 2025; Lai et al., 2024). Implicit PRMs derive step-level signals from outcome feedback alone (Yuan et al., 2025; Snell et al., 2024), yet recent benchmarks expose persistent weaknesses on fine-grained error categories and inter-step consistency (Song et al., 2025; Zheng et al., 2025). A shared limitation is that scoring remains step-local, treating each step as conditioned only on its immediate predecessor. *TopoPRM trains no neural verifier and claims no proof-level guarantee, instead introducing deterministic global-structure supervision over dependency topology together with local continuity.*

On-Policy Distillation and Reasoning Compression. Knowledge distillation transfers capability from larger teachers (Hinton et al., 2015), and on-policy variants close the train-test gap by supervising student-generated trajectories (Gu et al., 2024; Agarwal et al., 2024; Ko et al., 2024; Song and Zheng, 2026). Reasoning-oriented OPD includes OPSDC (Sang et al., 2026), ExOPD (Yang et al., 2026), BOND (Sessa et al., 2024), and KDRL (Xu et al., 2025), with black-box GAD broadening applicability beyond logits access (Ye et al., 2025); in parallel, long-CoT compression has been pursued through token pruning, length penalties, and explicit budget control (Xia et al., 2025; Luo et al., 2025; Arora and Zanette, 2026; Aggarwal and Welleck, 2025; Muennighoff et al., 2025). Most of these objectives still optimize token-level matching without explicit structural constraints, even as interpretability studies reveal graph-like latent organization in long-CoT reasoning (Yao et al., 2023; Besta et al., 2024; Zhong et al., 2026). *TopoPRM drives both topology-aware reward optimization and on-policy self-distillation, so that compression is conditioned on structural adequacy rather than token imitation alone.*

3 Method

TGSD (Figure 2) is a topology-aware post-training framework whose three stages share a **DAG Scorer**

$\mathcal{E}$ that converts reasoning traces into structure-sensitive representations. Stage I performs a supervised cold start to enforce parsable formatting. Stage II runs GRPO with a hierarchical reward that fuses TopoPRM with ACE. Stage III executes topology-guided self-distillation, in which the Stage-II teacher provides token-level reverse-KL supervision while the student rolls out under topology-conditioned revision prompts.

3.1 Problem Setup

Let $x = (q, c)$ denote an input pairing a mathematical problem q with optional context c , and let π_θ be a policy that generates a reasoning trace $y = (s_1, \dots, s_T) \sim \pi_\theta(\cdot | x)$, where each s_i is a textual reasoning step. Outcome-only RLVR supervises y with a sparse final-answer reward $r_{\text{out}}(x, y) \in [0, 1]$. We define $\mathcal{G}_y = (\mathcal{V}, \mathcal{A})$ as a directed dependency graph on y , with node set $\mathcal{V} = \{s_1, \dots, s_T\}$ and edge set $\mathcal{A} \subseteq \mathcal{V} \times \mathcal{V}$ encoding surface-evidenced support relations between steps, and $\mathcal{E} : y \mapsto \mathcal{G}_y$ as the DAG Scorer that produces it. *Our objective is to recover latent dependency structure from $\mathcal{G}_y$ as unified supervision across SFT, RL, and distillation, without human step labels or learned verifier models.*

3.2 Topology-Aware Data Processing

Trace-to-DAG Extraction. Given a rollout $y = (s_1, \dots, s_T)$, the DAG Scorer $\mathcal{E}$ proceeds in three coupled stages whose outputs are reused by every downstream signal. **(i) Step segmentation and parsing.** $\mathcal{E}$ first splits y into reasoning steps using numbered markers, mathematical discourse markers, and line boundaries, then partitions the trace into sub-questions so that no edge can cross independent components. **(ii) Implicit feature extraction.** Each step s_i is reduced to three normalized feature sets: mathematical expressions E_i , canonicalized claim keys C_i , and variable references V_i , after information-density filters discard single symbols, pure numerals, and heading-like fragments that would otherwise inflate the matching surface. **(iii) DAG construction.** For every ordered intra-component pair (s_i, s_j) with $i < j$, $\mathcal{E}$ inserts an implicit edge $i \rightarrow j$ whenever $E_i \cap E_j$, $C_i \cap C_j$, or $V_i \cap V_j$ is non-empty; if s_j is a derivation or conclusion step still lacking incoming support, an *adaptive sequential weak* edge from s_{j-1} is added only when s_{j-1} does not already emit strong outgoing dependencies, which prevents the chain-dominant artefacts that long traces tend to produce. Full ex-

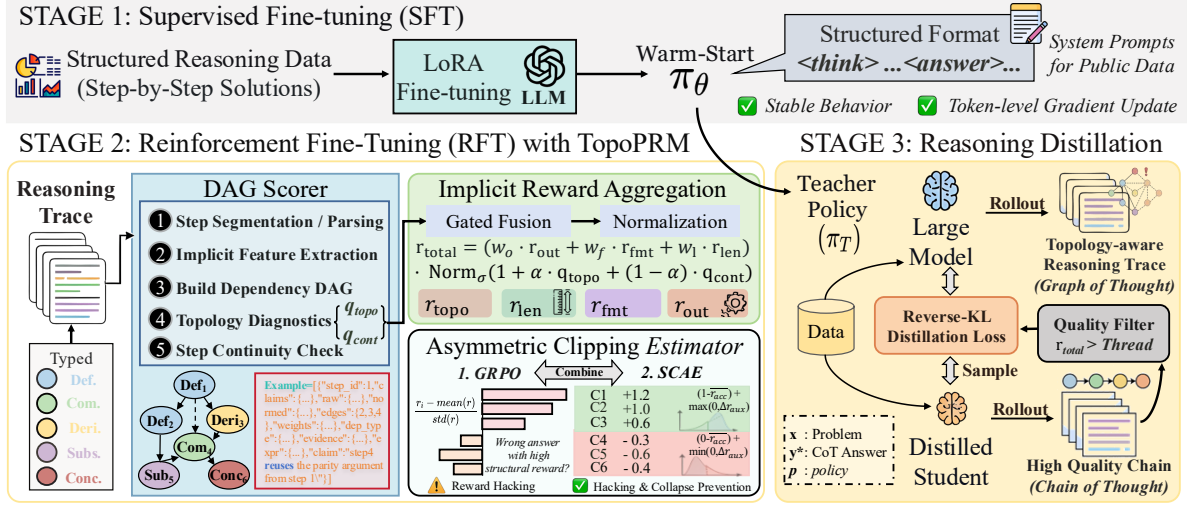


Figure 2: Overview of our proposed TopoPRM with post-training framework TGSD. The DAG Scorer lifts reasoning traces into structural representations. A hierarchical multiplicative reward and ACE bound within-stratum credit, while a topology-guided on-policy distillation stage compresses the long-CoT teacher into a compact student.

traction rules are deferred to Appendix C.

Topology and Continuity Diagnostics. On the resulting graph $\mathcal{G}_y = (\mathcal{V}, \mathcal{A})$ we compute two normalized scores that constitute the process-level reward signal. The *topology quality* $q_{\text{topo}} \in [0, 1]$ aggregates five structural checks, while the *continuity quality* $q_{\text{cont}} \in [0, 1]$ measures step-level traceability. The simplified formula is as follows:

$$q_{\text{topo}}(\mathcal{G}_y) = \mathbb{1}[\|\mathcal{V}\| > 0] + \mathbb{1}[\mathcal{G}_y \text{ acyclic}] + \mathbb{1}[\rho_{\text{orph}}(\mathcal{G}_y) = 0] + \delta(\mathcal{G}_y) + \kappa(\mathcal{G}_y, \mathcal{G}^*), \quad (1)$$

$$q_{\text{cont}}(y) = \gamma \eta(y) + (1 - \gamma) \mathbb{1}[\eta(y) = 1], \quad (2)$$

$$r_{\text{topo}}(y) = \alpha q_{\text{topo}}(\mathcal{G}_y) + (1 - \alpha) q_{\text{cont}}(y), \quad (3)$$

where $\gamma, \alpha \in (0, 1)$, ρ_{orph} , δ , and κ measure, respectively, the fraction of orphan conclusion nodes, the proportion of edges respecting the premise-to-consequence ordering, and the alignment with an optional reference DAG $\mathcal{G}^*$; $\eta(y) = \frac{1}{T} \sum_{i=1}^T \mathbb{1}[\text{supported}(s_i, s_{<i}, x)]$ is the fraction of reasoning steps in y whose entities overlap with the prefix $s_{<i}$ or the question x . Detailed formulas and optimizations are reported in Appendix A.1.

Hierarchical Aggregation. Weighted-sum rewards can misrank traces when high process scores compensate for incorrect final answers (Denison et al., 2024), and they tend to collapse the within-group variance on which GRPO relies (Yu et al., 2025; Kazemnejad et al., 2025). We therefore couple the correctness-based and structural signals multiplicatively, treating r_{topo} as a structural mod-

ulator on top of an outcome-anchored base:

$$r_{\text{total}}(x, y) = (w_o r_{\text{out}} + w_f r_{\text{fmt}} + w_l r_{\text{len}}) \cdot \text{Norm}_\sigma(1 + r_{\text{topo}}(y)), \quad (4)$$

where $r_{\text{out}} \in [0, 1]$ measures final-answer correctness, r_{fmt} scores parsability, and r_{len} regularises length. The structural modulator is wrapped by a batch-wise normalizer $\text{Norm}_\sigma(\cdot)$, and the multiplicative form preserves correctness primacy: a trace with $r_{\text{out}} = 0$ cannot be lifted by any process bonus, while traces with equal outcome scores remain distinguishable through their dependency structure. This yields the within-group reward dispersion that ACE exploits in Stage II.

3.3 Topology-Guided Post-Training

Stage I: Supervised Warm-Start. We perform Supervised Fine-Tuning (SFT) on structured mathematical reasoning data to serve as a warm-start that teaches the model to follow the required response format, produce stepwise solutions, and expose intermediate reasoning in a form parsable by $\mathcal{E}$. This reduces invalid rollouts during subsequent reinforcement learning and stabilizes reward computation, since topology diagnostics are meaningful only when the generated trace contains recoverable reasoning steps.

Stage II: GRPO with TopoPRM Rewards. Starting from the SFT checkpoint, we optimise the policy with GRPO under the TopoPRM reward. For each prompt, the policy samples a group of

completions, each of which is parsed into a dependency DAG by $\mathcal{E}$ and scored accordingly. The aggregated reward in Eq. 4 combines outcome correctness, format compliance, length regularization, and topology-derived process signals. Outcome rewards select for correct final answers, while topology and continuity rewards encourage the model to make intermediate dependencies explicit, avoid unsupported conclusions, and reduce structurally redundant reasoning.

Asymmetric Clipping Advantage Estimator (ACE). Multiplicative structural rewards introduce a potential hacking risk: a wrong response may appear well-structured under surface diagnostics. We propose ACE, an asymmetric clipping advantage estimator that fuses correctness-stratified credit assignment (Liu et al., 2026) with standard GRPO z-score clipping (Shao et al., 2024) into a single estimator,

$$\hat{A}_i = \begin{cases} \text{clip}(z_i^{\mathcal{C}} + \max(0, \Delta r_{\text{aux}}^{(i)}), 0, \bar{c}), i \in \mathcal{C}, \\ \text{clip}(z_i^{\mathcal{W}} + \min(0, \Delta r_{\text{aux}}^{(i)}), \underline{c}, 0), i \in \mathcal{W}, \end{cases} \quad (5)$$

where $\underline{c} < 0 < \bar{c}$ suppress tail noise within the assigned sign, $\mathcal{C}/\mathcal{W}$ are the correct/wrong strata defined by $r_{\text{out}}, z_i^{\pm} = (R_i - \mu_{\pm})/(\sigma_{\pm} + \epsilon)$ is the within-stratum z-score, and $\Delta r_{\text{aux}}^{(i)}$ is the deviation of r_{topo} from its stratum mean. The asymmetric max/min ensures structural signals can only re-rank traces *within* a correctness stratum, never flip its sign; the bounds $(\underline{c}, \bar{c})$ suppress tail noise. Together with batch-relative normalisation and a positive base floor for low-variance groups, this prevents reward collapse. Full derivation and the stable pseudo-code are deferred to Appendix B.

Stage III: Topology-Guided Self Distillation. The optimized model after Stage II serves as the teacher π_T , while a smaller pretrained checkpoint initializes the student π_θ . TGSD performs on-policy distillation conditioned on topology diagnostics. For each prompt x , the student first rolls out its own response $y^* \sim \pi_\theta(\cdot | x)$; the same extractor $\mathcal{E}$ produces $\mathcal{G}_{y^*}$ together with $(r_{\text{out}}, q_{\text{topo}}, q_{\text{cont}})$; a dispatcher ψ then issues a revision instruction $P_r = \psi(r_{\text{out}}, q_{\text{topo}}, k)$ targeting an orphan-conclusion position k in $\mathcal{G}_{y^*}$. The teacher then supervises the student under P_r , so that the signal concentrates on structurally suspect regions (orphan conclusions and continuity breaks) rather than spreading uniformly across the trace. Sampling y^* from π_θ rather

Algorithm 1 TGSD Training Pipeline

Require: Base policy π_0 ; data $\mathcal{D}_{\text{sft}}, \mathcal{D}_{\text{rl}}$; weights w_o, w_f, w_l, α

Ensure: Teacher π_T , compact student π_S

- 1: $\pi_\theta \leftarrow \text{SFT}(\pi_0, \mathcal{D}_{\text{sft}})$ ▷ Stage I
- 2: **for** each GRPO step on $\mathcal{D}_{\text{rl}}$ **do** ▷ Stage II
- 3: Sample $\{y^{(i)}\}_{i=1}^G \sim \pi_\theta(\cdot | x)$; build $\mathcal{G}^{(i)} = \mathcal{E}(y^{(i)})$
- 4: Compute $r_{\text{out}}^{(i)}, r_{\text{fmt}}^{(i)}, r_{\text{len}}^{(i)}, q_{\text{topo}}^{(i)}, q_{\text{cont}}^{(i)}$
- 5: $r_{\text{total}}^{(i)}$ by Eq. 4; $\hat{A}^{(i)}$ by Eq. 5
- 6: Update π'_θ with the GRPO objective
- 7: **end for**
- 8: $\pi_T \leftarrow \pi'_\theta$; init. π_S from a smaller pretrained checkpoint
- 9: **for** each prompt $x \in \mathcal{D}_{\text{rl}}$ **do** ▷ Stage III
- 10: $y_{\text{init}} \sim \pi_S(\cdot | x)$; compute $(r_{\text{out}}, q_{\text{topo}}, q_{\text{cont}})$ on $\mathcal{E}(y_{\text{init}})$
- 11: $P_r \leftarrow \psi(r_{\text{out}}, q_{\text{topo}}, k)$
- 12: $y_{\text{rev}} \sim \pi_T(\cdot | x, y_{\text{init}}, P_r)$; filter by format/topology
- 13: Update π_S with $\mathcal{L}_{\text{TGSD}}$ (Eq. 6)
- 14: **end for**
- 15: **return** (π_T, π_S)

than from a fixed teacher trace eliminates the train-test mismatch of off-policy distillation (Agarwal et al., 2024; Song and Zheng, 2026).

Distillation objective. Let $p_t \triangleq \pi_\theta(\cdot | x, y_{<t})$ and $q_t \triangleq \pi_T(\cdot | x, y^*, P_r, y_{<t})$ denote the student and teacher next-token distributions at position t , where $y_{<t}$ is the autoregressive prefix sampled on-policy by the student. Restricted to samples passing format and topology filters ($r_{\text{out}}^{(\text{rev})} = 1$ and $q_{\text{topo}}^{(\text{rev})} \geq \tau_g$), the student minimises

$$\mathcal{L}_{\text{TGSD}}(\theta) = \mathbb{E}_{x, y \sim \pi_\theta} \sum_{t=1}^{|y|} \mathcal{D}_{\text{RKL}}(p_t \| q_t). \quad (6)$$

$\mathcal{G}_{y^*}$ exposes broken support and P_r steers π_T toward the missing dependency, turning the per-token RKL into a structurally targeted gradient.

Critical dependency chain as the compression target. Equation (6) pushes the student toward a linearisation of $\mathcal{G}_{y^*}$ obtained by composing three operations on the DAG: *step-type contraction* merges consecutive same-type nodes; *layer aggregation* folds parallel topological generations into a layered chain $L_1 \rightarrow L_2 \rightarrow \dots$; and *transitive sparsification* replaces $\mathcal{G}$ by its transitive reduction $\text{tr}(\mathcal{G})$. The resulting *critical dependency chain*, analogous to the critical path in project scheduling, lower-bounds any structurally supported trajectory that closes the answer; the student is therefore trained to recover this minimal chain rather than to match teacher token frequencies, which is the structural counterpart of length-penalty compression. The full pipeline is summarised in Algorithm 1.

Table 1: Main results on nine public benchmarks across four task categories (Elementary: GSM8K, MATH-500; Olympiad: OlympiadBench, Omni-Math; Competition: AIME’24, AIME’25, CNMO’24; General: MMLU, GPQA-Diamond). All numbers are pass@5 accuracy (%) collected from the original papers or reproduced under the same evaluation protocol. For each backbone group, the best score per column is in bold and the second-best is underlined. Rows shaded in blue correspond to TopoPRM. Avg. is the unweighted arithmetic mean across all nine benchmarks.

Model	Params	Elementary		Olympiad		Competition			General			Avg.
		GSM8K	MATH	Olympiad	Omni	AIME'24	AIME'25	CNMO'24	MMLU	GPQA-D		
Reference open-source models (not bolded/underlined)												
Qwen3.5-2B	2B	77.2	60.8	48.5	13.3	6.7	13.3	36.1	32.8	37.9	36.3	
Qwen3.5-4B	4B	85.1	55.8	32.8	43.4	0.0	13.3	10.0	67.4	29.8	37.5	
Qwen2.5-Math-7B-Instruct (Yang et al., 2024b)	7B	95.2	73.8	38.0	22.0	13.0	10.0	13.3	54.8	28.3	38.7	
Llama-3.1-8B-Instruct (Grattafiori et al., 2024)	8B	84.5	52.2	15.0	10.0	3.0	2.0	3.0	68.0	30.0	29.7	
DeepSeek-R1-Distill-Llama-8B (DeepSeek-AI, 2025)	8B	83.0	89.1	48.0	25.0	50.4	35.0	25.0	52.0	49.0	50.7	
Qwen2.5-7B (Base) (Yang et al., 2024a)	7B	85.4	49.8	25.0	18.0	7.0	5.0	8.0	<u>74.2</u>	36.4	34.3	
+ SFT	7B	91.3	65.4	<u>42.5</u>	<u>54.2</u>	16.7	12.2	<u>16.7</u>	70.1	28.3	44.2	
+ SimpleRL-Zoo (Zeng et al., 2023)	7B	<u>91.7</u>	<u>78.2</u>	<u>38.6</u>	40.4	20.0	15.6	<u>15.0</u>	62.5	55.2	<u>46.4</u>	
+ Open-Reasoner-Zero (Hu et al., 2026)	7B	89.0	81.4	47.9	38.4	17.9	15.6	13.3	62.5	36.6	44.6	
+ TopoPRM (Ours) ★	7B	95.8	66.8	47.8	57.2	<u>16.7</u>	<u>12.2</u>	47.0	82.7	<u>51.0</u>	53.0	
DeepSeek-R1-Distill-Qwen-7B (Base) (DeepSeek-AI, 2025)	7B	73.8	36.8	25.8	28.0	46.7	30.0	22.9	23.7	12.6	33.4	
+ SFT	7B	86.0	92.8	52.0	48.0	<u>55.5</u>	40.0	30.0	50.0	49.1	<u>55.9</u>	
+ GRPO (Shao et al., 2024)	7B	<u>85.1</u>	67.4	<u>56.2</u>	<u>72.0</u>	50.0	<u>40.0</u>	57.8	<u>50.1</u>	17.7	55.1	
+ TopoPRM (Ours) ★	7B	84.3	<u>66.6</u>	56.4	72.2	55.5	40.0	<u>55.4</u>	60.3	<u>34.3</u>	58.3	
Qwen3.5-9B (Base)	9B	69.5	66.6	55.2	61.5	16.7	21.1	16.7	28.6	37.4	41.5	
+ SFT	9B	<u>94.1</u>	<u>68.2</u>	44.0	53.8	30.0	15.6	30.0	<u>78.4</u>	47.0	51.2	
+ GRPO (Shao et al., 2024)	9B	93.3	63.0	<u>56.2</u>	72.0	46.7	30.0	57.8	63.0	17.7	<u>55.5</u>	
+ TopoPRM (Ours) ★	9B	97.9	81.9	56.2	<u>55.7</u>	<u>43.3</u>	<u>18.9</u>	<u>47.0</u>	86.4	<u>40.4</u>	58.6	

4 Experiments

4.1 Setup

We conduct extensive experiments to evaluate our methods on nine benchmarks spanning mathematical reasoning and general tasks: GSM8K, MATH-500, Olympiad, Omni-MATH, AIME 2024, AIME 2025, CNMO 2024, MMLU, GPQA-D, comparing against more than ten baselines including reference open-source models. Among them, we adopt Qwen3.5-9B, Qwen2.5-7B (Yang et al., 2024a), and DeepSeek-R1-Distill-Qwen-7B (DeepSeek-AI, 2025) as the primary base models. Training variants use: (i) LoRA SFT (von Werra et al., 2022; Hu et al., 2022) on $\sim 19.5K$ DAG-annotated math traces, with 3 epochs, 64 rank, learning rate 5×10^{-6} and $\alpha=128$. GRPO is run for 300 steps, 4 generations/prompt with learning rate 5×10^{-6} , $\beta=0.04$, equipped with outcome-only reward or the hierarchical TopoPRM reward (Eq. 4). The hybrid topology extractor combines rule templates with batched LLM dependency inference and is invoked asynchronously alongside rollout generation (cost analysis in §4.4). The primary metric is accuracy, while pass@k / maj@5 / token-efficiency are recorded for completeness (Wang et al., 2023). Full setting and implementation details are reported in Appendix A.

4.2 Main Results (RQ1)

Unified Accuracy Evaluation. Table 1 reports pass@5 accuracy across nine benchmarks for three backbone families. TopoPRM achieves the highest average accuracy (53.0%, 58.3%, and 58.6%) among all training variants on the three base models, respectively. On DeepSeek-R1-Distill-7B, TopoPRM matches or exceeds the GRPO baseline on 7 of 9 benchmarks and improves MMLU by +10.2 points (60.3 vs. 50.1), indicating that topology supervision transfers to general reasoning without task-specific tuning. On Qwen2.5-7B, TopoPRM surpasses the strong SimpleRL-Zoo baseline on 5 of 9 benchmarks (Olympiad: 47.8 vs. 38.6; Omni: 57.2 vs. 40.4; MMLU: 82.7 vs. 62.5) while remaining competitive elsewhere.

Shorter Chains, but Higher Efficiency. Table 2 further decomposes performance by token cost. TopoPRM (Full) achieves the highest Acc/kTok on all four primary benchmarks (GSM8K: 118.2, MATH-500: 35.1, OlympiadBench: 18.6, Omni-MATH: 26.6), generating 15–24% fewer tokens than outcome-only GRPO while maintaining equal or higher accuracy. This pattern—higher accuracy at lower token cost—is the signature of structurally tighter reasoning chains, where topology supervision prunes redundant derivation branches.

Table 2: Per-benchmark results and component ablations of TOPOPRM. We report **p@1**, **p@5**, **maj@5**, mean generated tokens / total wall-clock seconds (**Tok / Time**, in ks), and token-efficiency **Acc/kTok** = $p@1 / (\text{tokens} / 1000)$. We use the same protocol in A. **Bold / underline**: best / second-best per column.

Dataset/Method	GSM8K (1,319)						MATH-500 (500)						OlympiadBench (134)						Omni-MATH (262)					
	p@1	p@5	maj@5	Tok / Time	Acc / kTok		p@1	p@5	maj@5	Tok / Time	Acc / kTok		p@1	p@5	maj@5	Tok / Time	Acc / kTok		p@1	p@5	maj@5	Tok / Time	Acc / kTok	
Base (Qwen3.5-9B)	55.2	87.8	55.7	1536/45.0	35.9		19.8	39.2	21.4	1536/18.7	12.9		11.2	25.4	14.2	2560/26.5	4.4		16.4	33.2	17.9	2559/52.0	6.4	
Outcome Only	93.3	97.7	93.6	1047/34.2	89.1	<u>50.8</u>	59.0	55.2	1511/18.9	33.6		31.3	46.3	37.3	<u>2043/19.7</u>	<u>15.3</u>		41.6	<u>55.3</u>	48.9	<u>1845/35.2</u>	<u>22.5</u>		
w/o Topology	93.0	97.7	93.9	<u>946/25.0</u>	<u>98.3</u>	51.0	<u>59.4</u>	55.6	<u>1506/16.2</u>	<u>33.9</u>		29.8	<u>50.7</u>	<u>38.1</u>	<u>2468/11.8</u>	12.1		42.4	54.2	45.8	2430/17.8	17.4		
w/o Continuity	51.0	86.1	52.6	1528/48.3	33.4	21.6	39.6	23.2	1536/19.1	14.1		8.2	25.4	9.0	2560/ 8.7	3.2		14.9	30.5	17.6	2560/18.5	5.8		
w/o ACE	<u>93.8</u>	98.0	<u>94.6</u>	1006/28.3	93.2	<u>50.8</u>	<u>59.8</u>	<u>55.2</u>	1515/18.2	33.5		32.8	47.0	35.8	2491/ 9.6	13.2		42.0	53.8	47.3	2463/19.4	17.1		
★ TopoPRM (Full)	94.1	<u>97.9</u>	94.8	<u>796/19.7</u>	118.2	51.0	81.9	54.6	<u>1454/16.6</u>	35.1		32.8	50.7	38.8	<u>1768/18.1</u>	18.6		42.8	55.7	<u>47.7</u>	<u>1592/19.0</u>	26.9		

Dataset/Method	AIME 2024 (30)						AIME 2025 (90)						CNMO 2024 (30)						MMLU (500)					
	p@1	p@5	maj@5	Tok / Time	Acc / kTok		p@1	p@5	maj@5	Tok / Time	Acc / kTok		p@1	p@5	maj@5	Tok / Time	Acc / kTok		p@1	p@5	maj@5	Tok / Time	Acc / kTok	
Base (Qwen3.5-9B)	16.7	33.3	33.3	4083/ 8.5	4.1		2.2	4.4	1.1	2560/ 5.9	0.9		10.0	16.7	3.3	2560/ 1.9	3.9		28.6	41.4	26.2	512/55.1	55.9	
Outcome Only	16.7	33.3	20.0	2546/ 5.6	6.6	13.3	15.6	8.9	2547/16.7	<u>5.2</u>		16.7	33.3	<u>26.7</u>	<u>2530/ 5.5</u>	6.6		63.0	84.0	61.0	512/146.2	123.0		
w/o Topology	<u>20.0</u>	<u>40.0</u>	<u>30.0</u>	2560/ 2.1	<u>7.8</u>	12.2	<u>16.7</u>	10.0	2560/ 6.2	4.8		<u>20.0</u>	40.0	<u>26.7</u>	2560/ 2.0	<u>7.8</u>		<u>65.8</u>	<u>85.0</u>	<u>63.0</u>	512/32.3	<u>128.5</u>		
w/o Continuity	6.7	13.3	3.3	2560/ 2.3	2.6	1.1	4.4	0.0	2560/ 7.0	0.4		6.7	20.0	3.3	2560/ 2.4	2.6		20.2	34.8	18.6	512/80.4	39.5		
w/o ACE	<u>20.0</u>	30.0	23.3	2560/ 2.6	<u>7.8</u>	<u>13.3</u>	<u>17.8</u>	12.2	2560/ 7.7	<u>5.2</u>		<u>20.0</u>	30.0	<u>26.7</u>	2560/ 2.6	<u>7.8</u>		61.1	83.1	61.2	508/58.1	120.3		
★ TopoPRM (Full)	30.0	43.3	33.3	<u>2431/ 5.4</u>	12.3	15.6	18.9	<u>11.1</u>	<u>2457/ 7.7</u>	6.4		30.0	47.0	30.0	<u>2530/ 2.5</u>	11.9		68.2	86.4	68.8	<u>508/48.0</u>	134.2		

Table 3: Reward component ablation on Qwen3.5-9B. Avg is the unweighted mean pass@1 across 8 benchmarks. Δ is the drop relative to the full model. Acc/kTok = Avg / (mean tokens / 1000). Collapse% is the fraction of GRPO rollout groups with near-zero reward std (from training logs).

Reward Config.	Avg (%)	Δ	Tok	Acc/kTok $\uparrow$	Collapse% $\downarrow$
TopoPRM (Full)	45.6	—	1692	26.9	37.9
w/o ACE	41.7	−3.8	1958	21.3	42.1
w/o Topology	41.8	−3.8	1943	21.5	44.6
w/o Continuity	16.3	−29.3	2047	8.0	68.8
Outcome Only	40.8	−4.7	1823	22.4	57.8

4.3 Ablation Studies (RQ2)

Reward Component Contribution. Table 3 isolates the contribution of each reward component on Qwen3.5-9B across 8 public benchmarks. Removing continuity causes the largest degradation ($\Delta = -29.3$ points), with Collapse% rising from 37.9% to 68.8%, indicating that local step-traceability provides the primary gradient signal for GRPO. Removing topology or ACE each causes a moderate drop ($\Delta \approx -3.8$, Collapse% at 42–45%), while outcome-only GRPO shows a −4.7 point deficit with 57.8% collapse. The full model also achieves the shortest generation (1,692 tokens) and highest token-efficiency (26.9 Acc/kTok), suggesting that topology-aware supervision prunes structurally redundant branches rather than merely penalising length. At larger capacity (a pre-experiment used Qwen-32B in Appendix F), the dominant signal shifts from continuity to topology.

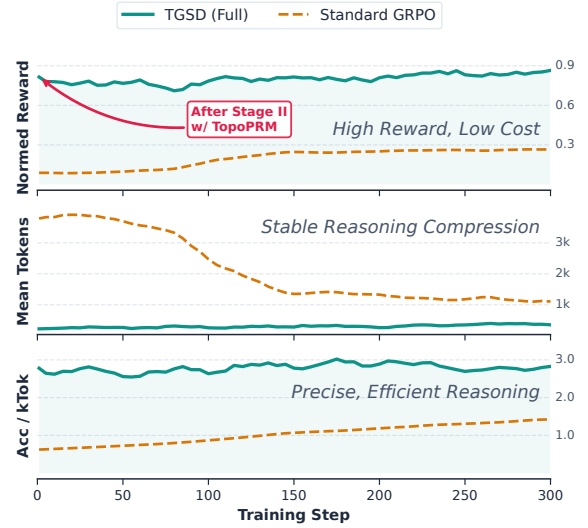


Figure 3: GRPO training dynamics over 300 gradient steps. Top: normalized reward trajectory. Middle: mean thinking trace length. Bottom: accuracy per thousand generated tokens, indicating effective of reasoning budget. TopoPRM converges with shorter completions and attains higher Acc/kTok, while the w/o-continuity variant inflates trace length with degraded token efficiency despite higher nominal reward.

Training Dynamics. Figure 3 visualizes the training metrics trajectory over 300 steps. Standard GRPO gradually compresses mean completion length from 3,700 to 1,200 tokens, whereas TopoPRM remains below 500 tokens throughout training—over $2.4\times$ shorter at convergence—while sitting on a high reward plateau from step 0 with strictly positive reward std. The w/o-continuity variant converges to a higher nominal re-

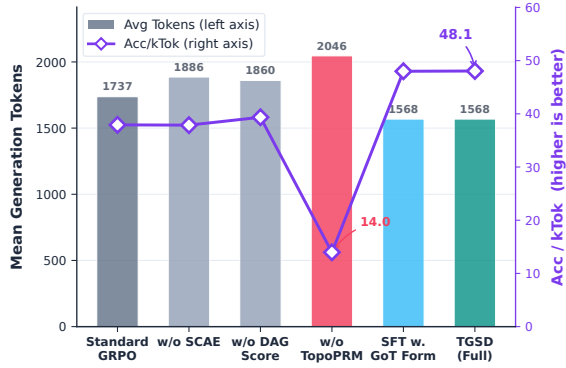


Figure 4: Token efficiency across reward configurations (Qwen3.5-9B, 4-benchmark average). Bars: mean generation tokens (left axis). Diamonds: Acc/kTok (right axis). TopoPRM (Full) achieves the best accuracy-per-token ratio with the shortest outputs.

ward but with near-zero reward std (bottom panel), indicating mode collapse: the model produces a single output pattern that scores well on the remaining reward components but lacks diversity for continued learning.

4.4 Efficiency Analysis (RQ3)

Accuracy per Token. Figure 4 summarizes the token-efficiency trade-off on four public benchmarks. TopoPRM (Full) achieves 48.1 Acc/kTok at 1,568 tokens, while outcome-only GRPO reaches 37.9 at 1,737 tokens, and the w/o-continuity variant collapses to 14.0 at 2,046 tokens. The monotonic degradation confirms that topology and continuity contribute complementary efficiency gains: global structure prevents redundant branching, while local continuity prevents unsupported transitions that inflate length. A detailed analysis on our in-domain critique set is in Appendix F, Table 8.

Training Cost. Full TopoPRM post-training on $8 \times A100$ -80GB GPUs completes in ~ 30 GPU-hours, including SFT takes ~ 3 h, GRPO ~ 15 h, and TGSD ~ 12 h. DAG extraction runs on rule-based and LLM generation mixedly in parallel, adding negligible wall-clock overhead.

4.5 Cross-Scale Distillation (RQ4)

Topology-Guided Cross-Scale Distillation. Table 4 evaluates the TGSD stage, which uses the 9B TopoPRM teacher (Stage-II output) to distill cross-scale students via topology-guided on-policy distillation. The 4B student retains 98.2% of the teacher’s GSM8K accuracy (82.8 vs. 84.3) and 92.3% on MATH-500 (61.5 vs. 66.6) while reduc-

Table 4: Cross-scale distillation via TGSD. The 9B TopoPRM teacher (Stage-II output) is compressed into smaller students. All accuracy numbers are pass@1 (%). Token Ratio is the mean generation length relative to the teacher. Structural Retention aggregates Acyclic%, No-Orphan%, and Edge-Keep against the teacher DAG profile on MATH-500.

Scale	GSM8K	MATH	Tok	Ratio	Struct. Ret.
9B (Teacher)	84.3	66.6	679	1.00×	1.00×
4B	82.8	61.5	382	0.56×	0.93×
2B	78.9	57.1	305	0.45×	0.88×
0.8B	73.6	52.4	244	0.36×	0.81×
4B (SFT-distill)	79.4	58.2	438	0.64×	0.86×
4B (Off-policy KL)	76.8	49.3	4096	$>1.0\times$	0.41×

ing token usage to $0.56\times$ and preserving 93% structural retention (acyclicity, edge coverage). Scaling down to 2B and 0.8B yields graceful degradation: accuracy drops 5–11 points per halving of parameters, while structural retention remains above 81%. The SFT-distill baseline at 4B achieves lower accuracy (79.4 GSM8K) with worse structural retention ($0.86\times$) despite generating longer outputs (438 vs. 382 tokens), confirming that topology-guided on-policy sampling produces a higher-quality training signal than static teacher traces. The off-policy KL baseline is generated at the token cap (4096) with only $0.41\times$ structural retention, leaving the student with near-zero usable supervision (details in Appendix G). Cross-scale ablation results on our in-domain validation set (Appendix F) further show that topology rewards are most effective at larger model scales (32B: $\Delta = -13.0$ for outcome-only), where the model has sufficient capacity to produce structurally coherent DAGs.

5 Conclusion

We presented TopoPRM, a topology-aware process reward model, together with its post-training framework TGSD, which recasts long reasoning traces as latent dependency DAGs and reuses a single set of structural diagnostics for both reward shaping and reasoning compression. A hybrid graph extractor produces implicit process rewards, a correctness-first hierarchical aggregation bounds structural credit through ACE, and TGSD distills the critical dependency chain as its compression objective. These components address failure modes that outcome-only supervision cannot distinguish, while requiring no human step labels, no learned verifier, and no graph-format decoding.

Limitations

TopoPRM provides a structural process signal rather than a proof-level verifier. Consequently, locally plausible yet semantically wrong steps can still receive positive structural credit, capping the headroom of topology-aware rewards on adversarial benchmarks such as AIME and CNMO where a single invalid deduction decides the outcome. We acknowledge that on competition-level benchmarks (AIME/CNMO) outcome-only GRPO can outperform TopoPRM at 9B, suggesting the structural reward trades sharp single-derivation accuracy for broader generalization. This trade-off is expected: topology diagnostics reward *structural soundness* (well-supported dependency chains) rather than *semantic correctness* of individual deduction steps, and the two objectives can diverge on problems where a single creative leap determines the answer. Three directions may close this structural–semantic gap. First, the hybrid extractor can be tightened by training a lightweight dependency model on traces with verified edges, so that rule templates and LLM inference are jointly calibrated against ground-truth logical support. Second, coupling topology-aware rewards with a generative process verifier would let structural credit pass only when a semantic check on the step is satisfied, which we expect to lift accuracy on adversarial sets while preserving token-efficiency gains. Third, extending TopoPRM beyond mathematics to code, scientific, and multimodal reasoning would allow executable or simulator-based oracles to supply the semantic signal that the current framework lacks.

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A Implementation Details

A.1 Hyperparameters

Table 5 lists the key hyperparameters for each training stage.

Table 5: Hyperparameter summary for the three training stages.

Parameter	Value	Description
<i>Stage I: Supervised Fine-Tuning (SFT)</i>		
Base model	Qwen3.5-9B	–
Adapter	LoRA $r=64$, $\alpha=128$	all-linear modules
Learning rate	5×10^{-6}	cosine, 3% warmup
Epochs	3	–
Batch size	2/device, accum 8	effective batch 16
Max seq length	8192	–
Precision	bf16	–
<i>Stage II: GRPO with TopoPRM Rewards</i>		
Base model	SFT checkpoint	–
Adapter	LoRA $r=64$, $\alpha=128$	all-linear modules
Learning rate	5×10^{-6}	cosine, 5% warmup
Gradient steps	300	–
Batch size	1/device, accum 16	effective batch 16
Num generations	4 per prompt	for GRPO grouping
Temperature	0.8	rollout sampling
β (KL penalty)	0.04	–
Max seq length	4096	input + completion
Precision	bf16	–
<i>Stage III: Topology-Guided Self-Distillation (TGSD)</i>		
Teacher model	Stage-II (merged)	frozen during distillation
Student model	Qwen3-4B/2B/0.8B	–
Adapter	LoRA $r=64$, $\alpha=128$	all-linear modules
Learning rate	2×10^{-5}	cosine, 3% warmup
Epochs	1	–
Batch size	2/device, accum 4	effective batch 8
Student temperature	0.7	on-policy rollout
Teacher temperature	0.0	greedy logits
KL temperature	1.0	for reverse KL
Quality gate (q_{topo})	≥ 0.4	$\sim 35\text{--}50\%$ pass rate
Max seq length	4096	–
Precision	bf16	–

A.2 Reward Configuration

The hierarchical reward (Eq. 4) uses base-term weights $(w_o, w_f, w_l) = (0.70, 0.15, 0.15)$ for outcome, format, and length. The topology-continuity balance is $\alpha=0.6$. The base floor is 0.05 to prevent zero-variance groups when outcome is uniformly zero.

The gated reward variant (Table 3, “w/o ACE”) uses: $\delta=0.1$ (outcome-tier gap, constrained by $\delta < 1/6$), $\epsilon=0.5$ (process-quality influence bound within a tier), and topo-continuity mix weight 0.5. Batch-level min-max rescaling of q_{topo} and q_{cont} guarantees non-zero reward variance.

A.3 Hardware and Infrastructure

All experiments use $8 \times \text{NVIDIA A100-SXM4-80GB}$ GPUs. For GRPO, GPUs are partitioned into a 4-GPU vLLM rollout server (tensor-parallel) and a 4-GPU LoRA training group, coordinated via MS-Swift’s server-mode architecture (ModelScope Team, 2024) with NCCL broadcast to keep the rollout adapter current. The vLLM server uses 60% GPU memory utilization with a maximum model length of 8192 tokens.

A.4 Evaluation Protocol

All public benchmark evaluations use the same protocol: vLLM inference with temperature 0.0 (greedy) for pass@1, and temperature 0.7 with top- $p=0.95$ for pass@5/maj@5 sampling. Maximum generation length is set to 2560 tokens for competition benchmarks (AIME, CNMO, OlympiadBench, Omni-MATH) and 1536 tokens for elementary benchmarks (GSM8K, MATH-500). MMLU and GPQA-Diamond use a 512-token maximum. Answer extraction uses regex matching for boxed answers and JSON parsing for structured outputs.

B Justification of ACE Algorithm

We followed the stratified-clipping idea from PASC-GRPO (Liu et al., 2026) to estimate our multiplicative hierarchical reward (Eq. 4). The key modifications are (i) replacing the auxiliary reward r_{aux} with our normalized topology/continuity diagnostics q_{topo} and q_{cont} from the extracted DAG, and (ii) fusing the stratified clip with standard GRPO z-score clipping into the unified form of Eq. (5) (**the two are equivalent through** $z_i^\pm \approx (R_i - r_{\text{acc}})/\sigma_\pm$). Algorithm 2 provides the numerically stable implementation of Stratified Clipping Advantage Estimator (SCAE) from PASC-GRPO. Our modified asymmetric clipping ensures that structural signals can only re-rank traces within a correctness stratum, never across it.

Algorithm 2 Stratified Clipping Advantage Estimator (SCAE)

Require: Group of completions $\{c^{(1)}, \dots, c^{(G)}\}$ for prompt q

Require: Outcome rewards $\{R_{\text{out}}^{(i)}\}_{i=1}^G$, auxiliary rewards $\{r_{\text{aux}}^{(i)}\}_{i=1}^G$

```

1:  $\bar{r}_{\text{acc}} \leftarrow \frac{1}{G} \sum_{i=1}^G R_{\text{out}}^{(i)}$ 
2:  $\mathcal{C} \leftarrow \{i : R_{\text{out}}^{(i)} = 1\}$  ▷ correct set
3:  $\mathcal{W} \leftarrow \{j : R_{\text{out}}^{(j)} = 0\}$  ▷ incorrect set
4:  $\bar{r}_{\text{aux}}^+ \leftarrow \text{mean}(\{r_{\text{aux}}^{(i)} : i \in \mathcal{C}\})$  if  $\mathcal{C} \neq \emptyset$  else 0
5:  $\bar{r}_{\text{aux}}^- \leftarrow \text{mean}(\{r_{\text{aux}}^{(j)} : j \in \mathcal{W}\})$  if  $\mathcal{W} \neq \emptyset$  else 0
6: for  $i \in \mathcal{C}$  do
7:    $A^{(i)} \leftarrow (1 - \bar{r}_{\text{acc}}) + \max(0, r_{\text{aux}}^{(i)} - \bar{r}_{\text{aux}}^+)$ 
8: end for
9: for  $j \in \mathcal{W}$  do
10:   $A^{(j)} \leftarrow -\bar{r}_{\text{acc}} + \min(0, r_{\text{aux}}^{(j)} - \bar{r}_{\text{aux}}^-)$ 
11: end for return Advantages  $\{A^{(i)}\}_{i=1}^G$ 

```

C DAG Extraction Details

Extraction pipeline. Given a reasoning trace in `<think>`, we apply four training-free stages:

1. Step segmentation: split by numbered-step markers, mathematical discourse markers (`:`, `;`, `.`), and line boundaries.
2. Sub-question partitioning: markers such as “Sub-question k ” define connected components; cross-sub-question edges are disallowed.
3. Information extraction: for each step, extract expressions, claims, and variables with rule-based patterns.
4. Edge construction: add dependency edges from explicit evidence (expression reuse, claim-key overlap, variable reference), then add adaptive weak sequential edges only when strong evidence is absent.

Information-density filters. To avoid low-information nodes and noisy links: (a) claims are extracted at sentence level, with heading fragments and unfinished prefixes removed; (b) trivial tokens (single symbols, pure numerals, non-relational fragments) are excluded from dependency matching.

Dependency rules. For each pair of steps (r_i, r_j) with $i < j$ in the same sub-question, we add an edge if: (1) r_j contains an informative expression first introduced in r_i ; (2) canonicalized claim keys overlap; or (3) r_j reuses variables introduced in r_i . When a derivation/conclusion step has no explicit support, a fallback edge from the previous step is inserted.

Adaptive sequential weak edges. A weak sequential edge (weight 0.3) is added only when the target step lacks strong incoming support and the source step does not already emit strong outgoing dependencies. This prevents chain-dominant artifacts in long traces.

Node typing. Each node is classified into DEFINITION, DERIVATION, COMPUTATION, CONCLUSION, AUXILIARY, SUBSTITUTION, or CASE_ANALYSIS by keyword rules.

Relationship between DAG quality and reasoning compression. We observe empirically that traces with higher DAG scores (acyclicity, low orphan ratio, strong directional consistency) tend to admit shorter compressed forms under TGSD. Intuitively, a well-structured DAG has fewer redundant branches and shorter critical paths, so the transitive reduction yields a more compact chain. The topology reward r_{topo} used during GRPO training (Eq. 3) is designed to push the model toward producing such compressible structures, while the DAG score itself serves as a diagnostic metric rather than a direct training signal.

D Reward Function Details

Outcome reward. Parses the model’s <answer> JSON to extract the predicted score and critique conclusion. Exact score match yields 1.0; within ± 1 yields 0.5. Matching the conclusion adds 0.5 bonus. The total is capped and normalized to $[0, 1]$.

Format reward. Checks for <think>...</think> and <answer>...</answer> blocks with valid JSON: both present $\rightarrow 1.0$; only valid <answer> $\rightarrow 0.5$; only valid <think> $\rightarrow 0.1$; otherwise $\rightarrow 0.0$.

Length reward. Linear decay from 1.0 (at $\leq 2,000$ characters) to 0.0 (at $\geq 4,000$ characters).

Topology reward (r_{topo}). The topology reward used as a training signal in GRPO is computed from the DAG quality score q_{topo} (Eq. 7). It aggregates five binary/continuous indicators: valid graph existence, acyclicity, absence of orphan conclusions, dependency-direction consistency, and reference-edge alignment (when reference DAGs are available).

Continuity reward. Uses regex-based expression matching to verify that each step references expressions or claims from prior steps or the problem statement. The continuity score q_{cont} (Eq. 8) measures the fraction of supported steps.

E Supplementary Mathematical Definitions

Topology score (full form). The topology-quality score used in Eq. 4 is

$$q_{\text{topo}}(\mathcal{G}) = \sigma(\lambda_b \mathbb{I}[|\mathcal{V}| > 0]) + \lambda_a \mathbb{I}[\mathcal{G} \text{ acyclic}] + \lambda_o \mathbb{I}[\rho_{\text{orphan}}(\mathcal{G}) = 0] + \lambda_d \delta(\mathcal{G}) + \lambda_k \kappa(\mathcal{G}, \mathcal{G}^*), \quad (7)$$

where $\sigma(\cdot)$ denotes the activation and normalized function, ρ_{orphan} is orphan-conclusion ratio, δ is dependency-direction consistency, and κ is reference-edge alignment (optional when reference DAGs exist). The five indicators in Eq. (7) are aggregated with weights $(\lambda_b, \lambda_a, \lambda_o, \lambda_d, \lambda_k) = (0.1, 0.2, 0.3, 0.2, 0.2)$ satisfying $\sum_i \lambda_i = 1$, which guarantees $q_{\text{topo}} \in [0, 1]$ and is consistent with the $[0, 1]$ normalization assumed in Eq. (3). The weights are not uniform because different DAG properties contribute unequally to a faithful simplification of the latent reasoning graph: *orphan-conclusion absence* ($\lambda_o = 0.3$) is weighted highest because an unsupported terminal node directly invalidates the chain; *acyclicity* and *direction-consistency* ($\lambda_a = \lambda_d = 0.2$) preserve the partial order required for a valid linearisation; *reference-edge alignment* ($\lambda_k = 0.2$) anchors the trace to ground-truth dependency when available; and *graph existence* ($\lambda_b = 0.1$) is a cheap sanity check that contributes the least information once the other four pass.

Continuity score (full form). The local continuity quality is

$$q_{\text{cont}}(y) = \begin{cases} 1.0, & \eta(y) = 1, \\ 0.8 \eta(y), & \eta(y) < 1, \end{cases} \quad \eta(y) = \frac{1}{T} \sum_{i=1}^T \mathbb{I}[\text{supported}(s_i, s_{<i}, q)]. \quad (8)$$

Remark. Eq. (8) is the engineering-tuned form used in our codebase, equivalent to Eq. (2) with $\gamma=0.8$: when $\eta(y)=1$ both reduce to 1.0, and when $\eta(y)<1$ both reduce to $0.8\eta(y)$.

Correctness-first stratification. For each group, we partition samples as

$$\mathcal{C} = \{i \mid r_{\text{out}}^{(i)} \geq \tau\}, \quad \mathcal{W} = \{i \mid r_{\text{out}}^{(i)} < \tau\}. \quad (9)$$

Within each stratum, we standardize and asymmetrically clip:

$$\tilde{r}^{(i)} = \begin{cases} \text{clip}(z^{(i)}, 0, 1.5), & i \in \mathcal{C}, \\ \text{clip}(z^{(i)}, -1.5, 0), & i \in \mathcal{W}. \end{cases} \quad (10)$$

Topology-conditioned revision dispatch. The revision instruction is generated by

$$P_r = \psi(r_{\text{out}}, q_{\text{topo}}, k), \quad (11)$$

where k denotes an orphan-conclusion position from the extracted DAG.

F Preliminary Experiments on In-Domain Data

F.1 Origin: Teacher Grading as an Implicit PRM Signal

Our work originated from access to an online teacher-grading platform for Chinese middle-school and high-school mathematics, where professional teachers score student solutions and leave step-level written critiques. Each record contains a problem, the student’s reasoning, per-step teacher annotations, and a final rubric score.

These data have two properties relevant to reasoning post-training. First, teacher critiques are step-localized: they mark which intermediate step introduces an error or which derivation is logically disconnected. Second, the grading rubric is topology-aware: teachers penalize solutions that reach a correct answer through a broken chain, and reward solutions with clean dependency structure even when the answer is partially wrong. This observation led to the design of TopoPRM: instead of training a learned PRM to judge each step, we extract a dependency DAG structure and let its topology serve as a free, verifier-free step-level signal.

The teacher-critique dataset serves two purposes: (a) it grounds the failure modes we target (dependency omission, spurious linearization, branch redundancy); (b) it provides a high-signal in-domain validation set for fine-grained ablations. All main results are reported on public benchmarks (§4); the teacher-critique set is used only for the ablations below.

F.2 Dataset and Protocol

The validation set contains 7,595 items (2,181 middle-school, 5,414 high-school), each with problem text, student response, per-step teacher verdict, and an integer rubric score. The model outputs a JSON blob with a critique conclusion and rubric score; Acc denotes exact-match on the rubric score, Fmt% denotes JSON compliance, and Len is mean response length in tokens. All GRPO runs start from the same SFT checkpoint, train for 200 steps with the hyperparameters in Table 5, and are evaluated under the unified protocol.

Table 6: Main results on the in-domain mathematical critique benchmark. **Middle** and **High** denote middle-school and high-school subsets. Acc measures critique scoring accuracy; Fmt% measures structural compliance; Len reports mean prediction length in tokens. Best results per column are in **bold**, second-best are underlined.

Model	Middle School			High School			Overall
	Acc	Fmt%	Len	Acc	Fmt%	Len	
Zero-shot Baseline							
Qwen3-32B	16.0	64.5	1 200	7.0	43.5	1 240	11.5
Qwen3-32B + SFT	29.0	78.5	1 164	19.5	65.0	1 233	24.3
RFT Variants							
+ Outcome Only	19.5	89.0	410	13.0	85.5	475	16.3
+ w/o Topology	23.0	91.0	397	16.5	86.0	460	19.8
+ w/o Continuity	27.5	90.5	379	20.0	85.0	436	23.8
+ Clipped Linear	29.0	93.0	382	21.5	88.5	458	25.3
+ TopoPRM (full)	37.2	94.6	364	22.3	89.3	424	29.7
TGSD Fine-tuned							
Qwen3.5-9B + full	40.5	98.0	1 602	30.5	94.0	2 191	35.5
Qwen3-8B + full	60.0	94.0	532	58.0	97.0	613	59.0

Observation 1: correctness-first shaping unlocks PRM-style gains. Table 6 shows that the full TopoPRM reaches 29.7% overall critique accuracy (37.2% middle, 22.3% high), corresponding to +13.4 points over outcome-only GRPO and +9.9 over the topology-ablated variant. TopoPRM simultaneously achieves the highest format compliance and the shortest responses (364/424 tokens), consistent with the hypothesis that once correctness primacy is enforced via ACE, the topology/continuity diagnostics act as a free step-level PRM signal.

Table 7: Reward component ablation on **Qwen3-32B** over the in-domain critique validation set. Each configuration toggles the topology (r_{topo}) and continuity (r_{cont}) signals on top of the outcome reward r_{out} . **Avg** is the unweighted mean of Middle- and High-school accuracy; Δ denotes the gap relative to the full TopoPRM. Rows are sorted by Avg in ascending order to highlight the monotonic contribution of each component.

Configuration	Reward Components			Accuracy (%)			
	r_{out}	r_{topo}	r_{cont}	Mid	High	Avg	Δ
Outcome Only	✓	✗	✗	19.5	13.0	16.3	−13.0
w/o Topology	✓	✗	✓	23.0	16.5	19.8	− 9.5
w/o Continuity	✓	✓	✗	27.5	20.0	23.8	− 5.5
TopoPRM (Full)	✓	✓	✓	33.0	25.5	29.3	—

Observation 2: each reward component contributes positively at adequate capacity. Table 7 isolates the contribution of every component of the TopoPRM reward on Qwen3-32B, the scale at which the generation cap rarely triggers, and the extracted DAG remains complete enough for the topology diagnostics to be well-calibrated. Three observations follow. (i) Removing the topology signal incurs the largest single-component drop ($\Delta = -9.5$), exceeding the loss from removing continuity ($\Delta = -5.5$), which indicates that *global dependency structure* dominates supervision once final-answer correctness is enforced. (ii) The two structural signals are complementary rather than redundant: ablating either still leaves a substantial gap to the full configuration, so the multiplicative aggregation in Eq. 4 exploits both axes. (iii) Outcome-only RL trails the full TopoPRM by -13.0 points despite identical hyperparameters, confirming that Step-level structural shaping provides supervision orthogonal to the verifiable answer reward. We restrict this ablation to the 32B scale because on smaller backbones, the completion cap is hit on a non-trivial fraction of rollouts, yielding partially-extracted DAGs whose topology reward is miscalibrated; this is consistent with prior evidence that structural auxiliary rewards require sufficient model capacity to be reliable (Liu et al., 2026; Yu et al., 2025).

Table 8: Accuracy-length trade-off on the in-domain critique validation set (Qwen3-32B). Acc/kTok = Overall-Acc per 1,000 tokens. Len↑ is the length increase relative to TopoPRM (lower is better).

Config.	Middle		High		Efficiency	
	Acc	Len	Acc	Len	Acc/kTok↑	Len↑
TopoPRM (full)	37.2	364	22.3	424	75.5	—
w/o Continuity	27.5	379	20.0	436	58.4	+4.1%
w/o Topology	23.0	397	16.5	460	46.2	+9.0%
Outcome Only	19.5	410	13.0	475	36.8	+12.6%
Qwen3-32B (base)	16.0	1200	7.0	1240	9.4	+230%

Observation 3: structural shaping compresses responses. Table 8 analyses the accuracy-length trade-off. TopoPRM achieves 75.5 accuracy points per 1,000 tokens, the highest token efficiency, while keeping responses 11.2% shorter than outcome-only GRPO (when extended to public benchmarks, this compression ratio further enlarges to 15%–24%). Removing topology or continuity monotonically inflates response length and reduces accuracy, suggesting that the structural signal is a genuine step-level supervisor rather than a brevity regularizer.

Table 9: Reward aggregation strategy comparison. All variants use the same five reward components but differ in how signals are combined. Collapse% is the fraction of rollout groups with near-zero reward std during GRPO training.

Strategy	Mid	High	Fmt%	Len	Collapse%↓
Linear (weighted sum)	37.2	22.3	94.6	364	75.1
Clipped linear	29.0	21.5	93.0	382	67.7
Hierarchical (Ours)	37.2	22.3	94.6	364	37.9

Observation 4: multiplicative aggregation avoids reward collapse. Table 9 compares reward aggregation strategies. Naive additive aggregation leads to 75.1% reward collapse (GRPO batches with zero reward std); the hierarchical multiplicative formulation reduces this to 37.9%. In the teacher-critique regime where correctness is sparse, the reward signal must be structured hierarchically so that topology gains can only re-rank within a correctness stratum.

Table 10: Structural quality of extracted reasoning DAGs on the in-domain middle-school evaluation set. Acyclic% is the fraction of valid DAGs; No-Orphan% is the fraction of traces without unsupported conclusion nodes; Dir-Cons. is mean directional consistency; DAG Depth is mean dependency depth; Edge-Keep% is the fraction of dependency edges retained after compression.

Model	Acyclic%↑	No-Orphan%↑	Dir-Cons.↑	DAG Depth	Edge-Keep%↑
Qwen3-32B + SFT	100	61	0.49	1.05	43
+ GRPO (outcome only)	100	82	0.89	1.12	56
+ GRPO (w/o topology)	100	84	0.91	1.14	58
+ GRPO (w/o continuity)	100	83	0.91	1.13	57
+ GRPO (TopoPRM full)	100	89	1.00	1.17	60

Observation 5: downstream traces inherit cleaner structure. Table 10 reports DAG-level structural diagnostics. Models trained with the full TopoPRM reward produce traces with higher acyclicity rates, fewer orphan conclusions, and stronger directional consistency. Since these diagnostics are computed by the same extractor that drove training, the gains are internal-consistency evidence rather than an independent metric.

G TGSD Details

TGSD is an optional compression stage whose goal is to retain the teacher’s structural reasoning profile while reducing generation length in compact students.

Why pure reverse-KL distillation is insufficient. A naive off-policy distillation (Hinton et al., 2015; Gu et al., 2024) that minimises $\mathcal{L}_{\text{RKL}} = \mathbb{E}_{x \sim \mathcal{D}_f} D_{\text{KL}}(\pi_S(\cdot | x) \parallel \pi_T(\cdot | x))$ over filtered teacher traces can fail when teacher rollouts are long and often format-incomplete. In our pilot, only 0.4% of 32B teacher traces closed the required <answer> tag within the training budget, so the student did not learn a stable stopping pattern. This failure mode is the empirical origin of the reverse-KL row in Table 4 and motivates the two-phase TGSD recipe below.

Pipeline. We train compact students (4B/2B/0.8B) with the two-phase TGSD procedure summarised in Section 3.3: (i) topology-conditioned self-refinement, followed by (ii) topology-gated on-policy distillation. We keep only format-closed samples and log topology diagnostics (q_{topo} , q_{cont} , orphan ratio, acyclicity) for all retained traces.

Topology-conditioned self-refinement. Given an initial on-policy attempt y_{init} , we compute $(r_{\text{out}}, q_{\text{topo}}, q_{\text{cont}})$ and construct the revision prompt P_r (Eq. 11). We then sample $y_{\text{rev}} \sim \pi_{\theta}(\cdot | x, y_{\text{init}}, P_r)$ and keep the tuple only when the revised response is correct and format-closed. The refined model is trained with

$$\mathcal{L}_{\text{refine}} = -\mathbb{E} \log \pi_{\theta}(y_{\text{rev}} | x, y_{\text{init}}, P_r) - \mathbb{E} \log \pi_{\theta}([y_{\text{init}}, P_r, y_{\text{rev}}] | x). \quad (12)$$

Topology-gated on-policy distillation. For each problem, the student samples an on-policy response $y \sim \pi_{\theta}(\cdot | x)$, while the frozen refined model provides token-level targets conditioned on the same topology-aware prompt. The full objective is given in Eq. 6. The topology prompt focuses the distillation signal on structurally suspect regions (orphan conclusions, continuity breaks) while remaining independent of human step labels.

Structural views for distillation. The extractor exposes three composable views of $\mathcal{G}$: (a) step-type contraction (merge consecutive same-type nodes); (b) layer aggregation (fold parallel topological generations into a layered chain); (c) transitive sparsification (replace $\mathcal{G}$ by its transitive reduction). The composition yields a chain whose length upper-bounds the shortest reachability cover of $\mathcal{G}$, mirroring the critical-path intuition from project scheduling.